

STA 35C: Statistical Data Science III

Lecture 7: Qualitative Predictors & Potential Problems in Linear Regression

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Announcements

Homework 2 is due Thu, Oct 8, 11:59 PM PT

- Please submit on time and follow the submission instructions
 - To Gradescope via Canvas in a single PDF
 - Email submissions are not accepted

If you have questions or need help

- Instructor office hours: Wed, 1–2 PM (MSB 1143)
- Discussion: Thu, 11:00–11:50 AM (TLC 2212)
- TA office hours: Thu, 12–1 PM (MSB 1117)
- Feel free to ask questions during lecture, in office hours or post them on Piazza
 - Report in-class participation using the "[Participation in lectures](#)" survey on Canvas

* **After class:** Review the full slide decks and read relevant sections in the textbook

Agenda

Last time: Multiple linear regression

- Model, least squares, coefficient interpretation, model fit
- Motivation for higher-order and interaction terms

Today:

- Complete Lecture 6: polynomial regression, interactions, and overfitting
- Qualitative predictors via dummy variables
- Common pitfalls in linear regression

Recap: Confidence and prediction intervals

At fixed predictor values x , let $f(x) = \mathbb{E}[Y \mid X = x]$ and write $Y_{\text{new}} = f(x) + \varepsilon_{\text{new}}$

Two sources of prediction uncertainty:

$$Y_{\text{new}} - \hat{f}(x) = \underbrace{f(x) - \hat{f}(x)}_{\text{estimating the mean}} + \underbrace{\varepsilon_{\text{new}}}_{\text{individual noise}}$$

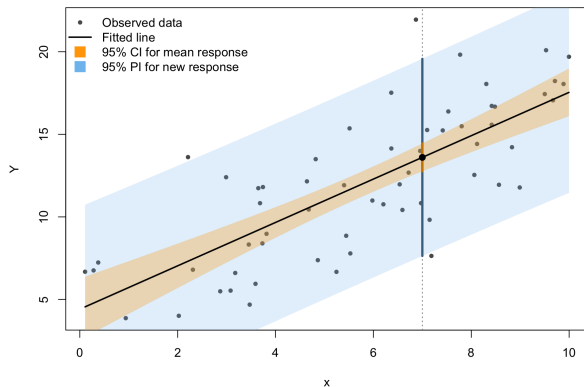
95% confidence interval for the mean $f(x)$:

- Accounts for uncertainty in estimating $f(x)$
- Over repeated training samples, 95% of these intervals cover the fixed mean $f(x)$

95% prediction interval for a new observation Y_{new} :

- Accounts for both estimation uncertainty and individual noise
- Over repeated training samples and independent new observations at the same x , 95% of these intervals cover Y_{new}

Confidence and prediction intervals: Illustration



At each fixed predictor value x_{new}

- Inner band: 95% CI for the mean response $E[Y | X = x_{\text{new}}]$
- Outer band: 95% PI for a new observation $Y_{\text{new}} | X = x_{\text{new}}$
- For the same fit, predictor value, and confidence level, the PI is wider because it includes individual noise

Exact formulas are beyond our scope, but in **R**, you can type:

```
predict(model, newdata = x0_df, interval = "confidence", level = 0.95)  
predict(model, newdata = x0_df, interval = "prediction", level = 0.95)
```

Completing Lecture 6: What about a non-linear relationship?

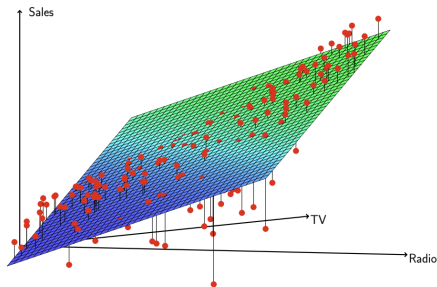


Figure: Pronounced synergy between **TV** and **Radio**; positive residuals cluster along the 45-degree line [JWHT21, Figure 3.5].

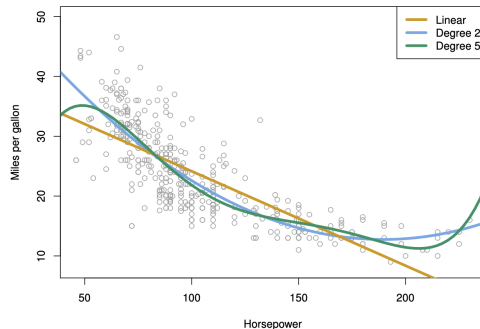


Figure: A non-linear relationship between **mpg** and **horsepower** is noticeable [JWHT21, Figure 3.8].

Linear model may not give you an adequate approximation

Completing Lecture 6: Polynomial regression

Polynomial regression extends the linear model by including powers of predictors¹:

$$Y = \beta_0 + \beta_1 X + \beta_2 X^2 + \cdots + \beta_d X^d + \varepsilon, \quad \mathbb{E}[\varepsilon \mid X] = 0$$

- Treated as multiple linear regression on transformed predictors $(X, X^2, \dots, X^d)$
- Fit by the least squares using $X, X^2, \dots, X^d$ as predictors
- The mean $\mathbb{E}[Y \mid X = x]$ can be nonlinear in x , while the model remains linear in its coefficients $\beta_0, \beta_1, \dots, \beta_d$

Example: Quadratic model

$$\text{mpg} = \beta_0 + \beta_1 \text{horsepower} + \beta_2 \underbrace{\text{horsepower}^2}_{\text{nonlinear term}} + \varepsilon$$

¹More generally, $Y = \sum_{\alpha: |\alpha| \leq d} \beta_{\alpha} \mathbf{X}^{\alpha} + \varepsilon$ where $\alpha = (\alpha_1, \dots, \alpha_p)$ and $\mathbf{X}^{\alpha} = X_1^{\alpha_1} X_2^{\alpha_2} \cdots X_p^{\alpha_p}$

Interaction terms: One slope depends on another predictor

Question: Can the association between **TV** and mean **Sales** depend on **Radio**?

Add an interaction term:

$$\begin{aligned}\text{Sales} &= \beta_0 + \beta_1 \text{TV} + \beta_2 \text{Radio} + \beta_3 \underbrace{\text{TV} \times \text{Radio}}_{\text{interaction term}} + \varepsilon \\ &= \beta_0 + (\beta_1 + \beta_3 \text{Radio}) \text{TV} + \beta_2 \text{Radio} + \varepsilon\end{aligned}$$

- Holding **Radio** fixed, the TV slope for mean sales is $\beta_1 + \beta_3 \text{Radio}$, not β_1 alone
- The product **TV** $\times$ **Radio** is another predictor; the model is still linear in its coefficients

Model flexibility and the risk of overfitting

A more flexible model can fit the training data more closely

- For least squares on the same observations, adding terms cannot increase training RSS^2

But a better fit on the training data does not necessarily mean a better model

- Our actual goal is good performance on new, unseen data
- A model that fits the training set extremely well may perform worse on a test set

Overfitting: If a model is too flexible, it can fit noise as well as signal

- Typical symptom: very small training error, but noticeably larger test error
- Compare training fit with performance on new data, not with training fit alone
- We will return to this issue repeatedly throughout the course

²Training $RSS = RSS$ measured on the training data

Pop-up quiz: Adding higher-order and interaction terms

Fit a multiple linear regression by least squares. Then add X_1^2 and X_1X_2 , retain the original predictors, and refit on the same observations.

Which statement is correct?

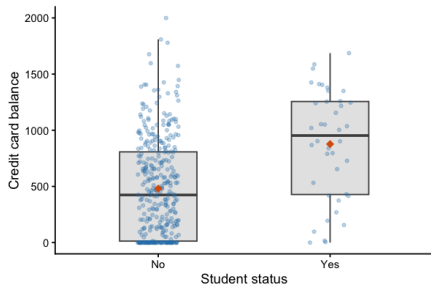
- A) Least squares no longer applies because the new terms are nonlinear in the original predictors.
- B) The fitted coefficients of the original predictors must remain unchanged.
- C) Training RSS cannot increase, but prediction error on new data may increase.
- D) Any decrease in training RSS guarantees better predictions on new data.

Answer: C. Setting the new coefficients to zero recovers the original fit; additional flexibility can also fit noise.

Qualitative predictors: Motivation

Motivating example: Credit dataset

- Response: **balance**
- Quantitative predictors: **age**, **cards**, **education**, **income**, **limit**, **rating**
- Qualitative (categorical) predictors: **own**, **student**, **married**, **region**
 - These do not have a natural numeric scale



Question: How do we incorporate categorical variables into a linear regression model?

- $\text{balance} = -0.4 \times \text{"own a house"} + 2.33 \times \text{"not a student"} - \dots ?$
- We cannot put labels like Yes/No or East/West/South directly into a linear model

Answer: Use “indicator (dummy) variables” to numerically encode each categorical level

Dummy variables

Idea: Represent each category by indicator (dummy) variables.

Two levels: For each binary variable, create a dummy variable:

$$D = \begin{cases} 1 & \text{if Yes} \\ 0 & \text{if No} \end{cases}$$

Then use D as a predictor in regression:

$$Y = \beta_0 + \beta_1 D + \cdots + \epsilon$$

- Example: Homeowner status $\text{own} \in \{\text{Yes}, \text{No}\}$

K levels ($K \geq 2$): With an intercept, use $K - 1$ dummies and one baseline level

Dummy variables: Illustration

Example

We encode $\text{region} \in \{\text{East, West, South}\}$ using dummy variables

Region	D_{West}	D_{South}
East	0	0
West	1	0
South	0	1

- East: $\mathbb{E}[Y \mid \text{East}] = \beta_0$
- West: $\mathbb{E}[Y \mid \text{West}] = \beta_0 + \beta_1$
- South: $\mathbb{E}[Y \mid \text{South}] = \beta_0 + \beta_2$

$$Y = \beta_0 + \beta_1 D_{\text{West}} + \beta_2 D_{\text{South}} + \varepsilon, \quad \mathbb{E}[\varepsilon \mid \text{region}] = 0$$

- The intercept β_0 is the baseline-group mean
- Each dummy coefficient β_1, β_2 is a difference from the baseline

Why $K - 1$ dummies?

- Naively coding East=1, West=2, South=3 would impose a spurious order/spacing
- If we also included D_{East} , then the intercept is redundant

Interpretation of dummy variable coefficient

Simple linear regression with a dummy ($\mathbb{E}[\epsilon \mid D] = 0$):

$$Y = \beta_0 + \beta_1 D + \epsilon, \quad \text{where } D \in \{0, 1\}.$$

- If $D = 0$: $Y = \beta_0 + \epsilon$.
- If $D = 1$: $Y = (\beta_0 + \beta_1) + \epsilon$.
- β_1 : The *difference* between the two group means ($D = 1$ vs. $D = 0$)

Under the usual inference assumptions, a t -test compares the two population means:

- $H_0 : \beta_1 = 0 \implies \text{no difference}$
- $H_1 : \beta_1 \neq 0 \implies \text{different population means}$

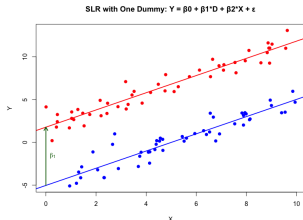
Dummy variables with a quantitative predictor

Assume $\mathbb{E}[\varepsilon \mid D, X] = 0$ and write $m_d(x) = \mathbb{E}[Y \mid D = d, X = x]$

No interaction:

$$Y = \beta_0 + \beta_1 D + \beta_2 X + \varepsilon$$

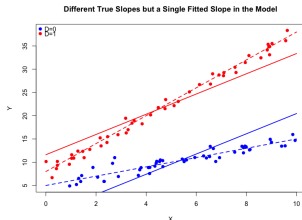
- $D = 0 \rightarrow m_0(x) = \beta_0 + \beta_2 x$
- $D = 1 \rightarrow m_1(x) = (\beta_0 + \beta_1) + \beta_2 x$
- β_1 is a constant shift between two groups



With interaction:

$$Y = \beta_0 + \beta_1 D + \beta_2 X + \beta_3 (D \times X) + \varepsilon$$

- $D = 0 \rightarrow m_0(x) = \beta_0 + \beta_2 x$
- $D = 1 \rightarrow m_1(x) = (\beta_0 + \beta_1) + (\beta_2 + \beta_3)x$
- β_3 allows the slope to differ by group



$\rightarrow \beta_1$ reflects the *average* effect of D , *holding X fixed*; not necessarily a constant effect

Pop-up quiz: Dummy variables

Suppose that a predictor **region** has three levels:

$$\{\text{East, West, South}\}.$$

We fit the model

$$Y = \beta_0 + \beta_1 D_{\text{West}} + \beta_2 D_{\text{South}} + \varepsilon,$$

using East as the baseline, and obtain

$$\hat{\beta}_0 = 400, \quad \hat{\beta}_1 = 120, \quad \hat{\beta}_2 = -50.$$

Question: Which statement is correct?

- A) The predicted mean is 400 for East, 520 for West, and 350 for South.
- B) The predicted mean is 0 for East, 120 for West, and -50 for South.
- C) West is predicted to have 70 less than South.
- D) We should also include D_{East} to make the model complete.

Answer: A. The intercept is the baseline-group mean, and each dummy coefficient is a difference from that baseline.

Potential pitfalls in linear regression

Linear regression is powerful, but it can fail if certain assumptions are not met

Possible issues:

- Validity of model assumptions
 - Does the chosen linear model adequately describe the mean response of Y versus X ?
 - Are the errors ϵ_i truly uncorrelated?
 - Is the variance of ϵ constant?
- Outliers & High-leverage points
 - What if there are extremely unusual points in the training data?
- Collinearity among predictors
 - What if some predictors are highly correlated?

Let's examine what these problems entail, how to diagnose and possibly mitigate them

Pitfall 1: Nonlinear relationship

Issue: The mean response is not linear in the predictors

- Example: $Y \approx \beta_0 + \beta_1 X^2 + \epsilon$

Why problematic: A linear fit can be systematically biased (leading to a poor fit and large residuals)

How to diagnose: Residuals vs. fitted values show curvature or other systematic patterns (e.g., a systematic deviation from 0)

Possible remedies:

- Include nonlinear transformations of X
- Use a more flexible model

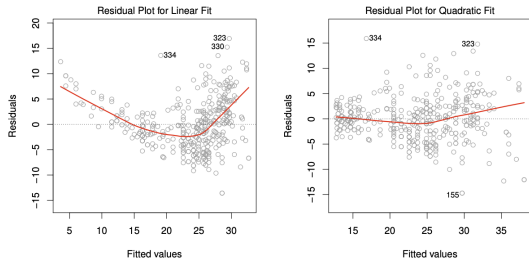


Figure: Plots of residuals vs. predicted values [JWHT21, Figure 3.9].

Pitfall 2: Correlated error terms

Issue: Errors $\{\epsilon_i\}$ are correlated, not independent

- Common in time series or grouped data (e.g., repeated measurements)
- If data is artificially duplicated or has a temporal pattern, errors can “track” each other

Why problematic: With a correct mean model, OLS remains unbiased; however, usual standard errors and inference can be *underestimated*

How to diagnose: Plot residuals in data order, by time, or by group; look for systematic patterns

Possible remedies:

- Use tailored methods for dependent data such as time series (ARIMA, etc.) or grouped data
- Carefully design experiments to avoid correlated errors

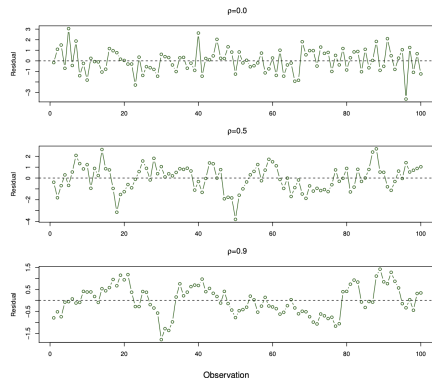


Figure: Plots of residuals from simulated time series data [JWHT21, Figure 3.10].

Pitfall 3: Non-constant variance of the error term

Issue: The error variance is not constant (heteroskedastic)

- Standard assumption is $\text{Var}(\epsilon_i) = \sigma^2$ (constant)

Why problematic: OLS estimates are often still unbiased under a correct mean model, but standard errors and inference can be unreliable

How to diagnose: Residual plots show a funnel shape or changing spread across fitted values

Possible remedies:

- Transform the response ($\log Y$, $\sqrt{Y}$, etc.) to stabilize variance
- Use weighted least squares, or use heteroskedasticity-robust standard errors

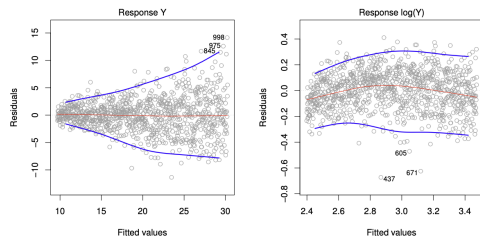


Figure: Residual plots with heteroskedastic error [JWHT21, Figure 3.11].

Pitfall 4: Outliers & high-leverage points

Issue: Influential points materially change the fitted model

- Outliers have unusual y -values
- High-leverage points have unusual x -values

Why problematic:

- Outliers can lead to a misfit, inflate RSE , and degrade R^2
- A small change in high-leverage points can pull the regression line substantially

How to diagnose:

- Residual plots, especially *studentized residuals*, can help identify outliers
- Leverage measures unusual x ; Cook's distance measures influence

Possible remedies: Inspect and possibly remove or adjust suspicious observations; use a “robust” statistical method

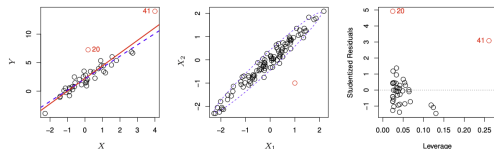


Figure: An illustration of outliers and high-leverage points [JWHT21, Figure 3.13].

Pitfall 5: Collinearity

Issue: Two or more predictors are highly correlated

- Example: $X_2 = X_1 + \text{small noise}$, or $X_3 = -2X_1 + 3X_2$, etc.

Why problematic:

- Difficult to separate individual effects
- Near collinearity causes *unstable* coefficients; exact collinearity makes them even non-unique

How to diagnose: Use variance inflation factors (VIFs); pairwise correlations can miss multivariable dependencies

Possible remedies:

- Drop one of the correlated predictors
- Combine or merge them (e.g., sum, average, or principal components)
- Use regularization techniques (e.g., ridge, lasso) to reduce variance

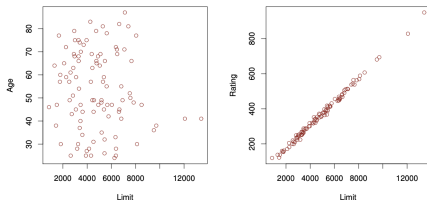


Figure: An illustration of high collinearity [JWHT21, Figure 3.14].

Wrap-up

- A mean-response confidence interval targets $\mathbb{E}[Y \mid X = x]$; a prediction interval targets a new observation
- Polynomial and interaction terms extend the mean model while remaining linear in the coefficients
- Qualitative predictors are incorporated through dummy variables; with an intercept and K levels, use $K - 1$ dummies
- Dummy coefficients are interpreted relative to the baseline group, and interactions allow group-specific slopes
- Linear regression can fail when assumptions break down, so residual plots and diagnostic tools are essential

Suggested reading: [JWHT21, Ch. 3.3; inference review: Ch. 3.1.2, 3.2.2]

Next lecture: Classification and logistic regression

References



Gareth James, Daniela Witten, Trevor Hastie, and Robert Tibshirani.

An Introduction to Statistical Learning: with Applications in R, volume 112 of *Springer Texts in Statistics*.

Springer, New York, NY, 2nd edition, 2021.